Identification of epimutations in rare diseases from a single patient perspective

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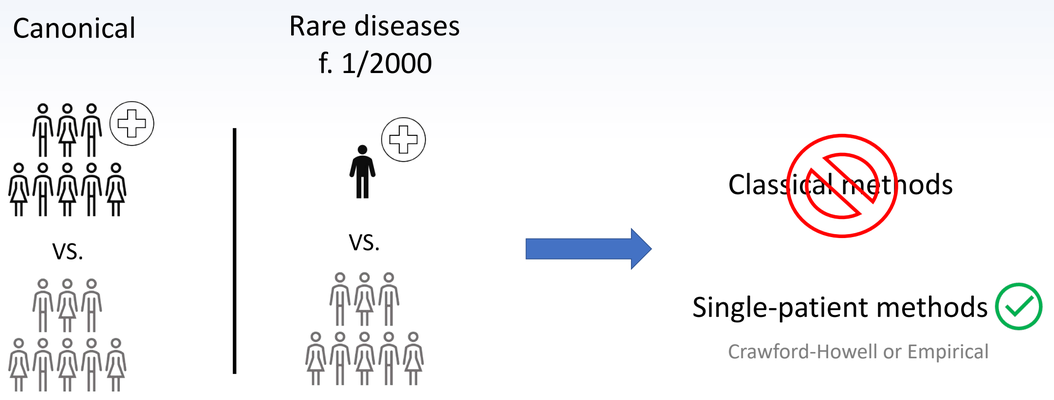
Background

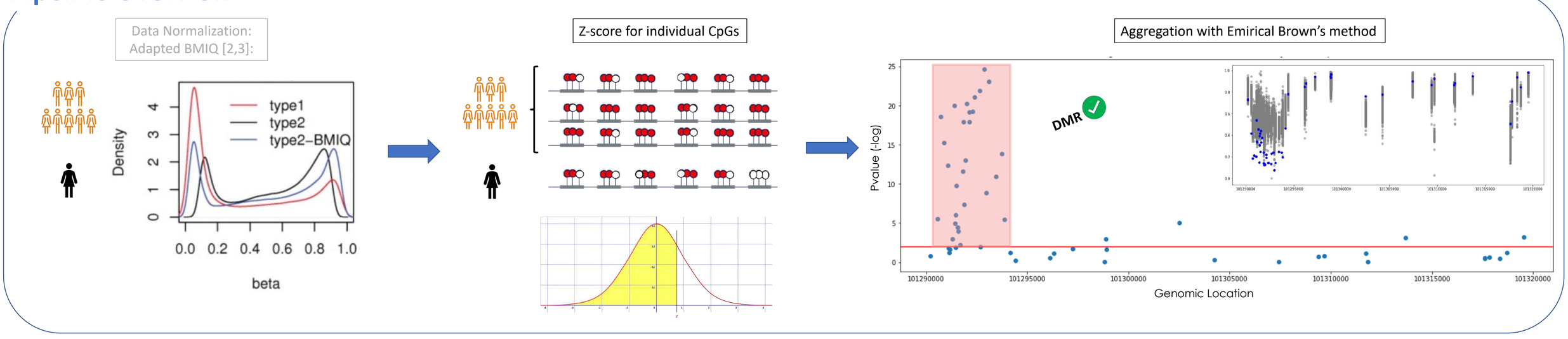
Canonical pipelines for the detection of epimutations based on methylation-array technologies rely on case-control group comparisons. However, in the context of rare diseases and multi-locus imprinting disturbances, **small cohorts** and **inter-patients' heterogeneity** prevent the use of those tools. Therefore, there is a need to provide a comprehensible and **statistically robust pipeline** for clinicians to perform analyses at the **single patient** level as well as characterize how different parameters may influence epimutation detection.



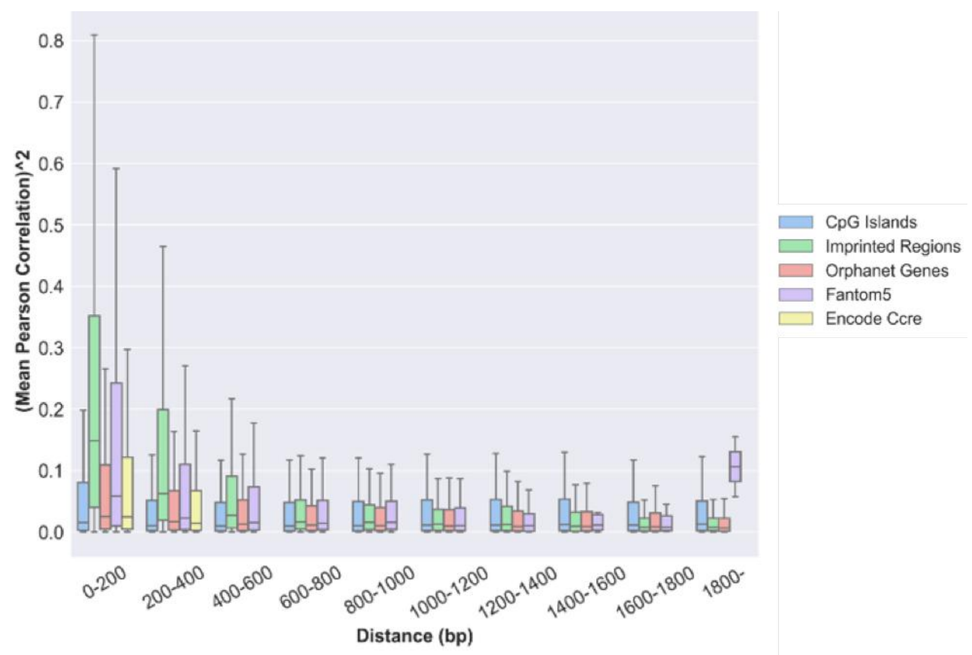
Results

We describe a statistical method to detect differentially methylated regions based on the **Z-score** and empirical **Brown method** from a single patient perspective. We generated **semi-simulated data** to assess how control population, effect size and region size affect epimutations detection, based on a public control population of 521 samples [1]. Finally, we **validated** the detection of methylation events in patients suffering **Beckwith–Wiedemann syndrome** (BWS) and rare **multilocus imprinting disturbance** (MLIDs) and discuss how this method is complementary to the process of clinical diagnosis.

Pipeline Overview

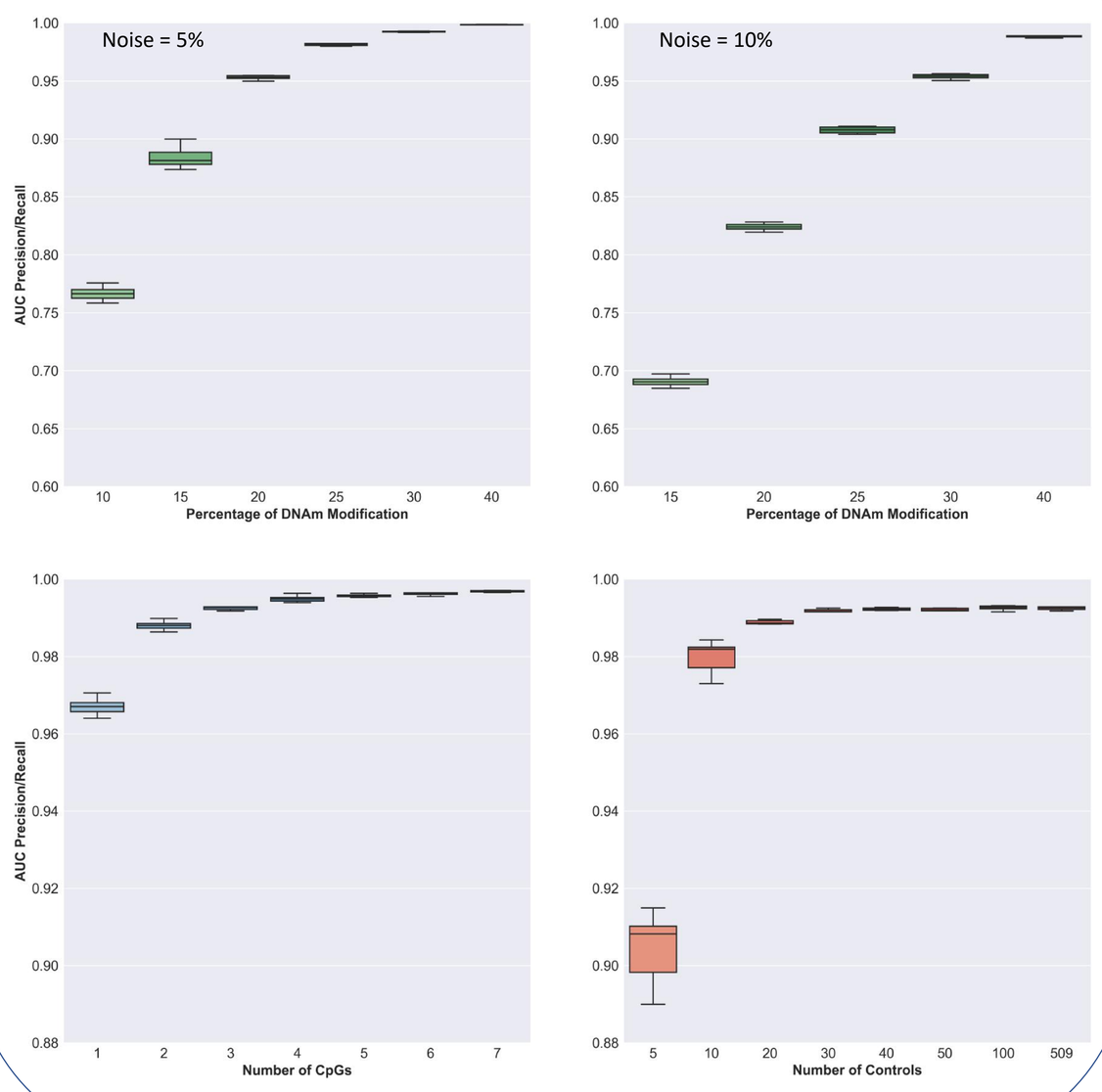


Correlation between CpGs in normal population



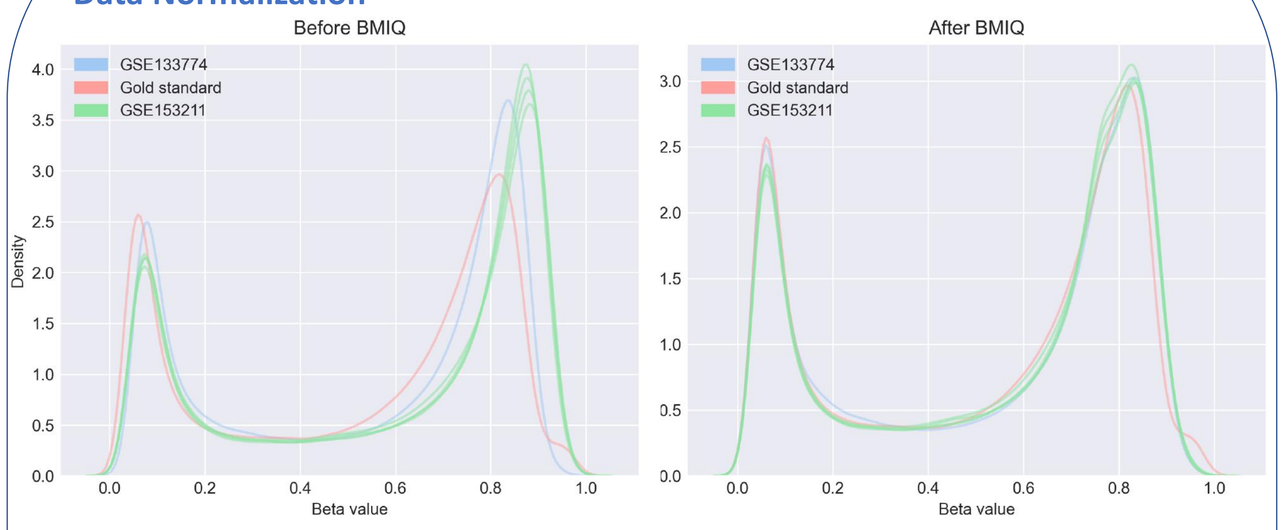
Take **dependence** into account when aggregating

Optima parameters for DMR identification

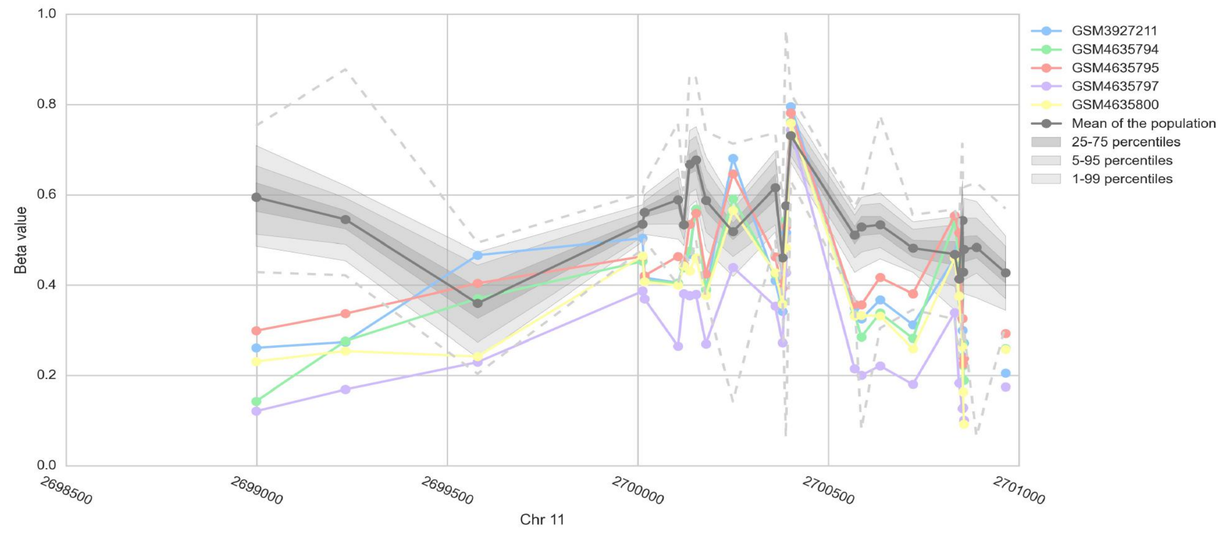


Detecting DMRs in Beckwith-Wiedemann patients

Data Normalization

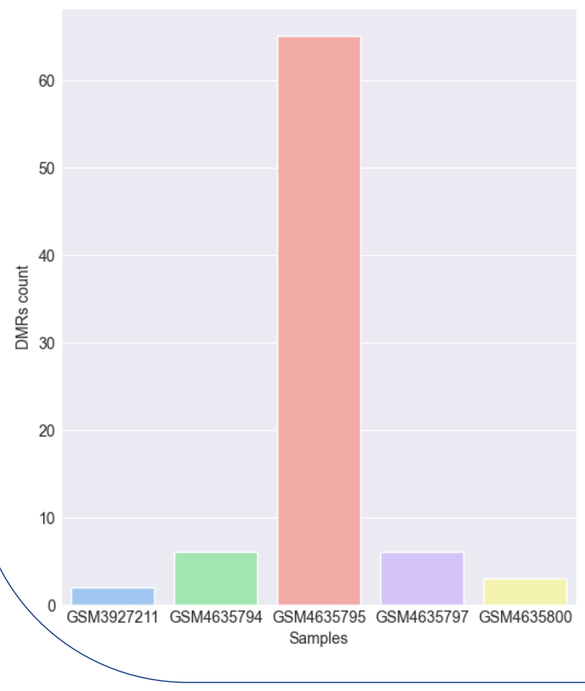


Cardinal locus of BWS patient



	GSM3927211	GSM4635794	GSM4635795	GSM4635797	GSM4635800
-log10(p-val)	9.709	15.678	4.23	60.319	20.475

Additional DMRs



Conclusion

We provided clinicians with a **robust** and **comprehensible method** to perform differential methylation analysis at the single patient level and showed its **diagnosis utility** in rare disorders.

References

1. Hannon E, Dempster EL, Mansell G, et al. DNA methylation meta-analysis reveals cellular alterations in psychosis and markers of treatment-resistant schizophrenia. Elife. 2021;10:e58430. Published 2021 Feb 26. doi:10.7554/eLife.58430
2. Teschendorff, Andrew E et al. "A beta-mixture quantile normalization method for correcting probe design bias in Illumina Infinium 450 k DNA methylation data." Bioinformatics (Oxford, England) vol. 29,2 (2013): 189-96. doi:10.1093/bioinformatics/bts680
3. Horvath, Steve. "DNA methylation age of human tissues and cell types." *Genome biology* vol. 14,10 (2013): R115. doi:10.1186/gb-2013-14-10-r115